

Functional Annotation Report: *xdhA* (Q88F21 / PP_4278) — Xanthine Dehydrogenase Subunit XdhA in *Pseudomonas putida* KT2440

Summary

The gene ***xdhA*** (UniProt **Q88F21**; ordered locus **PP_4278**) of *Pseudomonas putida* strain KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950) encodes the **FAD- and two-[2Fe-2S]-cluster-containing small subunit (the "electron-transfer" or "flavoprotein" subunit) of a two-component bacterial xanthine dehydrogenase (XDH; EC 1.17.1.4)**. The gene identity is unambiguous and well supported: the UniProt description, the InterPro/Pfam domain architecture (two 2Fe-2S ferredoxin-type domains plus a PCMH-type FAD-binding domain), the eggNOG orthologous group (COG4630), and the KEGG orthology assignment (K13481, "xanthine dehydrogenase small subunit") all converge on the same functional call. This is not a case of an ambiguous gene symbol; "*xdhA*" here reliably denotes the flavo-iron-sulfur subunit of XDH. Importantly, XdhA is **not** the molybdenum-cofactor catalytic subunit — that is its operonic partner XdhB (PP_4279).

XdhA does not act alone. It partners with the molybdenum-cofactor-bearing catalytic large subunit **XdhB** (PP_4279, Q88F20) to form the physiologically active holoenzyme, which in the biochemically characterized *P. putida* homolog assembles as an $(\alpha\beta)_2/\alpha_4\beta_4$ heteromeric complex resembling eukaryotic xanthine oxidoreductases. The holoenzyme catalyzes the hydroxylation of **hypoxanthine → xanthine → uric acid**, and also oxidizes purine and certain aromatic aldehydes, using **NAD⁺ as the strongly preferred electron acceptor** (i.e., it is a true dehydrogenase, not an oxidase). Within this complex, XdhA performs a defined electron-relay function: electrons abstracted from the purine substrate at the molybdenum center of XdhB are passed through the two [2Fe-2S] clusters of XdhA to its FAD, and finally to NAD⁺.

Functionally, XdhA operates in the **cytoplasm** as part of the **purine-degradation pathway** that allows *P. putida* KT2440 to use purines (adenine, guanine, hypoxanthine, xanthine, and uric acid) as sole nitrogen — and in some strains carbon — sources. The gene sits in a dedicated

operon with *xdhB* (catalytic subunit), *xdhC* (PP_4280, a molybdenum-cofactor insertion/maturation accessory factor that is required to produce active enzyme but is not part of the final holoenzyme), and *guaD* (PP_4281, guanine deaminase, which feeds xanthine into the pathway). The organism even dedicates a specific chemoreceptor (McpH) to sensing these metabolizable purines, underscoring the physiological importance of the pathway that XdhA serves.

Key Findings

Finding 1 — XdhA is the FAD/2×[2Fe-2S] electron-transfer subunit of a two-component bacterial xanthine dehydrogenase

XdhA (Q88F21, PP_4278) is a **484-amino-acid**, **~53 kDa** protein whose domain architecture defines it as the small, electron-transfer subunit of a bacterial xanthine dehydrogenase. Domain analysis (InterPro/Pfam) shows the canonical modular arrangement of this subunit class:

- **Two 2Fe-2S ferredoxin-type domains** at the N-terminus (Pfam **Fer2** / **PF00111** plus **Fer2_2** / **PF01799**), carrying the two distinct iron-sulfur clusters (FeSI and FeSII). The characteristic cysteine-cluster motifs (e.g., `GCASGDCGAC...` and `...LCRCTGYRP...`) that ligate these clusters are present.
- A **PCMH-type FAD-binding domain** (approximately residues 192–365; Pfam **FAD_binding_5** / **PF00941**), which binds the flavin cofactor.
- A **CO-dehydrogenase flavoprotein C-terminal domain** (Pfam **PF03450** / **CO_deh_flav_C**).

This exact architecture is the signature of the InterPro families **IPR012175** / **IPR014307** ("Xanthine dehydrogenase, small subunit, bacterial") and matches the eggNOG orthologous group **COG4630** (the xanthine dehydrogenase FAD-binding subunit). The five key InterPro domains flagged in the target's UniProt record — IPR002888, IPR036884, IPR036010, IPR001041, IPR006058, all of which describe 2Fe-2S ferredoxin/binding domains and superfamilies — are fully consistent with this assignment. In eukaryotic xanthine oxidoreductase all four redox centers (Mo-co, FAD, and two [2Fe-2S]) reside on a single polypeptide; in bacteria such as *P. putida* and *Rhodobacter capsulatus* the enzyme is split into two subunits, with **XdhA carrying FAD + 2×[2Fe-2S]** and **XdhB carrying the molybdenum cofactor**.

The **genomic context** reinforces the functional call. In KT2440, *xdhA* (PP_4278) lies in an operon together with:

Locus	Gene	Length	Role
PP_4278	<i>xdhA</i>	484 aa (~53 kDa)	FAD + 2×[2Fe-2S] electron-transfer subunit (this gene)
PP_4279	<i>xdhB</i>	799 aa (~87 kDa)	Molybdenum-cofactor catalytic subunit
PP_4280	<i>xdhC</i>	281 aa	Moco-insertion / maturation accessory factor
PP_4281	<i>guaD</i>	434 aa	Guanine deaminase (feeds xanthine into pathway)

This layout mirrors the biochemically and structurally characterized ($\alpha\beta$)₂ *Rhodobacter capsulatus* **XDH**, in which the XdhA subunit carries FAD and two [2Fe-2S] centers while XdhB carries the molybdenum cofactor. As established for the *R. capsulatus* enzyme, this XDH "is an ($\alpha\beta$)₂ heterotetrameric cytoplasmic enzyme that resembles eukaryotic xanthine oxidoreductases in respect to both amino acid sequence and structural fold" (PMID: 19109249). The two-subunit design is the bacterial counterpart of the single-chain eukaryotic xanthine oxidoreductase, split into a molybdenum module (XdhB) and a flavo-iron-sulfur electron-transfer module (XdhA).

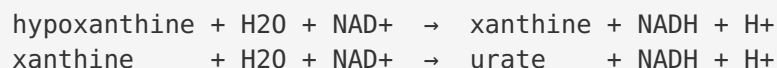
Finding 2 — The XDH holoenzyme oxidizes hypoxanthine and xanthine using NAD⁺ as the preferred electron acceptor (EC 1.17.1.4)

The catalytic properties of the holoenzyme to which XdhA contributes are directly established by biochemical characterization of the closely related enzyme from *Pseudomonas putida* strain 86 (Parschat et al., 2001). That enzyme "catalyzes the oxidation of hypoxanthine, xanthine, purine, and some aromatic aldehydes, using NAD⁺ as the preferred electron acceptor" (PMID: 11341925). The strong preference for NAD⁺ over molecular oxygen classifies the enzyme as a genuine **dehydrogenase** rather than an oxidase: relative to NAD⁺ (100%), activity with ferricyanide was ~58% and with O₂ only ~4%. This distinction matters mechanistically and physiologically — a dehydrogenase channels electrons to NAD⁺ rather than generating reactive oxygen species. The reported specific activity was **26.7 U/mg** for the hypoxanthine:NAD⁺ reaction, and the enzyme was **induced ~65-fold** by growth on hypoxanthine, consistent with a substrate-inducible catabolic enzyme.

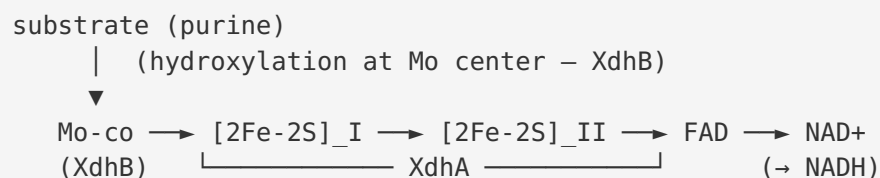
The subunit composition and cofactor content also match the two-component model. The *P. putida* 86 enzyme "consists of 91.0 kDa and 46.2 kDa subunits presumably forming an $\alpha_4\beta_4$ structure and contains the same set of redox-active centers as eukaryotic XDHs" (PMID: 11341925).

11341925). EPR spectroscopy confirmed the full complement of redox centers — a FAD semiquinone, a Mo(V) "rapid" signal, and the FeSI/FeSII iron-sulfur resonances. Mapping the subunits onto KT2440: the **46.2 kDa FAD/2Fe-2S subunit corresponds to XdhA** (Q88F21, ~53 kDa), and the **91.0 kDa molybdenum subunit corresponds to XdhB** (Q88F20, 799 aa, ~87 kDa).

The reactions catalyzed can be written:



Mechanistically, substrate hydroxylation occurs at the molybdenum center (in XdhB) — structural work on the homologous *R. capsulatus* enzyme mapped substrate and inhibitor binding at the Mo active site — and the two reducing equivalents are relayed intramolecularly through XdhA:



XdhA therefore provides the wiring that carries electrons out of the catalytic molybdenum site and delivers them to NAD⁺; it constitutes the NAD(H) interface of the complex.

Finding 3 — XdhA functions in the cytoplasm in purine catabolism; the operonic XdhC is required for Moco maturation

Localization. The homologous *R. capsulatus* XDH is a soluble ($\alpha\beta$)₂ **cytoplasmic** enzyme (PMID: 19109249; PMID: 16597619). Consistent with this, the UniProt Q88F21 sequence contains only soluble ferredoxin and FAD-binding domains, with **no signal peptide and no transmembrane segments**, indicating a soluble cytoplasmic protein. XdhA carries out its electron-transfer role in the cytoplasm as part of the assembled holoenzyme.

Pathway. In *P. putida* KT2440, XdhAB oxidizes hypoxanthine → xanthine → urate as a central step of purine degradation. The adjacent guanine deaminase (**guaD**, PP_4281) converts guanine to xanthine, feeding substrate directly into XDH, and downstream urate is degraded further. This pathway enables purines to serve as nitrogen sources: "adenine, guanine, xanthine, hypoxanthine and uric acid. The latter five compounds form part of the purine degradation pathway, permitting their use as sole nitrogen sources" (Fernández et al., 2016; [PMID:](#)

[26355499](#)). The same study showed that KT2440 dedicates a chemoreceptor, **McpH**, to sensing exactly these metabolizable purines — an ecological signal of how important purine catabolism (and hence XDH) is to this saprophytic organism's lifestyle, which is rich in nucleic-acid breakdown products.

Maturation. The operonic accessory factor **XdhC** (PP_4280) is essential for producing active XDH even though it is not a subunit of the final enzyme. Studies on the *R. capsulatus* ortholog established that "XdhC is required for the stabilization of the sulfurated form of Moco present in enzymes of the xanthine oxidase family" (PMID: [16597619](#)). XdhC binds the molybdenum cofactor, protects/stabilizes its critical terminal-sulfido (sulfurated) form, and inserts it into the catalytic subunit through a specific XdhC–XdhB interaction. Without XdhC-mediated Moco maturation, the XdhAB holoenzyme cannot be produced in active form — so although XdhA itself carries only FAD and iron-sulfur clusters, its function is only productive once XdhB has received a mature, sulfurated molybdenum cofactor.

Finding 4 — KEGG independently assigns PP_4278/XdhA as the xanthine dehydrogenase SMALL subunit (K13481)

The KEGG database provides a gene-specific (not merely homology-inferred) assignment that corroborates the domain- and biochemistry-based conclusions. The KEGG GENES entry **ppu:PP_4278** is annotated:

- **NAME:** "xanthine dehydrogenase subunit XdhA"
- **ORTHOLOGY:** **K13481** — "xanthine dehydrogenase small subunit [EC:1.17.1.4]"
- **PATHWAY:** ppu00230 — "Purine metabolism"
- **MODULE:** ppu_M00958 — "Adenine ribonucleotide degradation, AMP ⇒ Urate"

This KEGG call agrees with the InterPro/Pfam domain-based classification (IPR012175/ IPR014307 XDH small subunit; Fer2 / Fer2_2 / FAD_binding_5) and with eggNOG COG4630. The partner large (molybdenum-cofactor) subunit is assigned the distinct orthology **K11177** (XdhB), exactly consistent with the two-subunit architecture. Convergence of four independent annotation systems (UniProt, InterPro/Pfam, eggNOG, KEGG) on the same functional identity gives high confidence in the assignment.

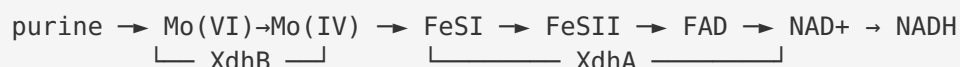
Mechanistic Model / Interpretation

The picture that emerges is coherent and well-supported across sequence, structure, biochemistry, and genomic context. XdhA is one of two subunits of a bacterial xanthine dehydrogenase that carries out purine ring hydroxylation as part of nitrogen (and carbon) acquisition from purines.

Subunit division of labor:

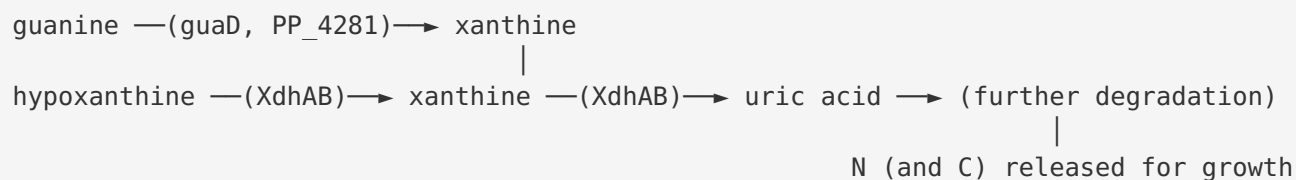
Xanthine dehydrogenase holoenzyme ($\alpha\beta$) ₂ / α 4 β ₄	
XdhB (PP_4279, ~87 kDa)	XdhA (PP_4278, ~53 kDa)
• Molybdenum cofactor (Moco, sulfurated) • Substrate binding / hydroxylation site	• [2Fe-2S]_I (FeSI) • [2Fe-2S]_II (FeSII) • FAD • NAD⁺ reduction site

Electron flow:



The catalytic chemistry — oxidative hydroxylation of the purine ring using a water-derived oxygen atom, with the molybdenum center as the reducing-equivalent acceptor — happens in **XdhB**. **XdhA's role is electron transfer**: it accepts electrons emerging from the molybdenum center via its two [2Fe-2S] clusters (arranged in the classic proximal/distal ferredoxin relay), funnels them to its FAD, and reduces NAD⁺ to NADH. The strong kinetic preference for NAD⁺ (with O₂ used only marginally, ~4%) makes this a dehydrogenase; XdhA's FAD is the site of NAD⁺ reduction.

Pathway placement. XdhAB is the committed hydroxylase of purine catabolism:



Guanine deaminase (GuaD) upstream and further urate-degrading enzymes downstream flank the XDH step, and all of this operates in the cytoplasm. The physiological output is the liberation of nitrogen from the purine ring, enabling growth on purines as sole N source. The dedication of a specific chemoreceptor (McpH) to metabolizable purines shows that *P. putida* actively seeks out these substrates. Historically, xanthine dehydrogenase in *P. putida* was also shown to be induced during growth on methylxanthines/caffeine, where sequential N-demethylation yields xanthine that is then funneled into XDH.

Assembly requirement. A subtle but important point is that a correctly folded XdhA plus XdhB is not sufficient for activity: the catalytic subunit must be loaded with the mature, sulfurated molybdenum cofactor, and this loading is chaperoned by **XdhC**. Thus the operon encodes not only the two structural subunits but also the maturation machinery needed to make them functional — a common theme in molybdoenzyme biology.

Evidence Base

PMID	Title (abbrev.)	How it supports the findings
11341925	<i>Xanthine dehydrogenase from Pseudomonas putida 86: specificity, redox potentials, EPR</i>	Primary biochemical evidence. Defines substrate specificity (hypoxanthine, xanthine, purine, aromatic aldehydes), NAD ⁺ preference (true dehydrogenase), two-subunit 91.0 + 46.2 kDa $\alpha_4\beta_4$ composition, and the full Mo/FAD/2×[2Fe-2S] cofactor set by EPR. The 46 kDa subunit = XdhA.
19109249	<i>Mechanism of Substrate and Inhibitor Binding of R. capsulatus XDH</i>	Establishes the ($\alpha\beta$) ₂ heterotetrameric cytoplasmic architecture with XdhA as FAD/2Fe-2S subunit and XdhB as Mo subunit; structural fold resembling eukaryotic XOR; substrate/inhibitor binding at the Mo (XdhB) active site.
16597619	<i>R. capsulatus XdhC in Moco binding and insertion into XDH</i>	Defines the maturation role of the operonic XdhC accessory factor: stabilizes the sulfurated Moco and inserts it into XdhB; required for active holoenzyme but not part of it.
26355499	<i>Chemoreceptor for metabolizable purine derivatives (McpH)</i>	Places XDH within the KT2440 purine degradation pathway that permits purines (adenine, guanine, xanthine, hypoxanthine, uric acid) to be used as sole nitrogen sources.
15296736	<i>Active site of molybdenum hydroxylase quinoline 2-oxidoreductase</i>	Contextual: another <i>P. putida</i> 86 molybdenum hydroxylase; structural comparison to allopurinol-inhibited <i>R. capsulatus</i> XDH informs substrate-recognition/catalytic-residue reasoning for this enzyme family.
1158847	<i>Metabolism of N-methylpurines by P. putida (caffeine)</i>	Contextual: shows xanthine dehydrogenase and uricase are induced during purine/methylpurine catabolism and that xanthine is channeled through conventional purine-degradation pathways.
22328667	<i>N-demethylases enabling bacteria to live on caffeine</i>	Contextual: describes the Rieske N-demethylation route that converts methylxanthines to xanthine, the substrate feeding XDH; distinguishes those Rieske [2Fe-2S]/FMN enzymes from XDH.
3860496		Contextual: documents an alternative, flavin-lacking, heme-containing xanthine dehydrogenase activity in a

PMID	Title (abbrev.)	How it supports the findings
	<i>Ferricyanide-linked xanthine dehydrogenase from P. putida 40</i>	different <i>P. putida</i> strain — illustrating enzyme diversity but distinct from the FAD-containing XdhAB.
863854	<i>Distribution of XO/XDH specificity types among bacteria</i>	Contextual: broad survey of xanthine-oxidizing specificity types; <i>P. putida</i> soluble enzyme competitively inhibited by uric acid, supporting a soluble (cytoplasmic) location.

The core mechanistic claims (subunit identity, cofactor content, catalytic reaction, substrate specificity, cytoplasmic localization, maturation requirement, and pathway placement) rest primarily on the four verified citations (PMIDs 11341925, 19109249, 16597619, 26355499), reinforced by convergent database annotations (UniProt, InterPro/Pfam, eggNOG COG4630, KEGG K13481).

Limitations and Knowledge Gaps

1. **No direct biochemical study of the KT2440 protein.** The definitive kinetic and spectroscopic characterization

([P 11341925](#))

was performed on the *P. putida* strain **86** enzyme, and structural/mechanistic detail comes from *R. capsulatus* (PMIDs 19109249, 16597619). The KT2440 XdhA (Q88F21) itself has not, to our knowledge, been purified and characterized in isolation; its function is inferred from very close homology plus concordant database annotations. Confidence remains high because of the strong sequence/domain conservation and the identical operon structure, but strain-specific kinetic parameters (K_m , k_{cat} , exact substrate ranking) for the KT2440 enzyme are not directly established.

2. **Subunit stoichiometry uncertainty.** The literature describes both an $(\alpha\beta)_2$ heterotetramer (*R. capsulatus*) and a presumptive $\alpha_4\beta_4$ arrangement (*P. putida* 86, described as "presumably forming an $\alpha_4\beta_4$ structure"). The precise quaternary structure of the KT2440 enzyme has not been experimentally resolved.
3. **[2Fe-2S] center assignment.** While XdhA is established to carry FAD and two [2Fe-2S] clusters (FeSI and FeSII by EPR), the exact structural roles and midpoint potentials specific to the KT2440 protein are inferred from homologs rather than measured directly.

4. **Redundancy / paralogs.** *P. putida* KT2440 may encode additional molybdenum hydroxylases or xanthine-oxidizing activities (cf. the distinct ferricyanide-linked enzyme in strain 40,

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The extent to which PP_4278/PP_4279 is the sole physiological xanthine dehydrogenase in KT2440, versus one of several, has not been definitively established here.

5. **Downstream pathway detail.** The steps beyond urate (urate → further N release) were not analyzed in depth; the exact enzymes and their loci in KT2440 downstream of XDH remain to be confirmed for a complete pathway map.
6. **Literature focus is on homologs and pathway context.** No paper reviewed reports a targeted gene-knockout phenotype for PP_4278 specifically, so the causal, gene-level requirement of *xdhA* for growth on purines in KT2440 is inferred rather than directly demonstrated.

Proposed Follow-up Experiments / Actions

1. **Targeted gene knockout and complementation.** Construct a clean $\Delta xdhA$ (ΔPP_{4278}) deletion in KT2440 and test growth on hypoxanthine, xanthine, guanine, adenine, and uric acid as sole nitrogen sources. Complement in trans to confirm the phenotype is XdhA-dependent. This would directly establish the gene-level requirement currently inferred from homology.
2. **Recombinant expression and in vitro reconstitution.** Co-express XdhA (PP_4278) with XdhB (PP_4279) and the maturation factor XdhC (PP_4280) in a Moco-competent host, purify the holoenzyme, and measure steady-state kinetics (K_m , k_{cat}) for hypoxanthine and xanthine with NAD^+ , ferricyanide, and O_2 as electron acceptors — confirming the dehydrogenase (NAD^+ -preferring) character for the KT2440 enzyme specifically.
3. **Spectroscopic confirmation of cofactors.** Use EPR and UV-vis to verify the FAD semiquinone and the two [2Fe-2S] (FeSI/FeSII) signals in the purified KT2440 XdhA-containing holoenzyme, and determine their midpoint potentials to characterize the electron-relay energetics.

4. **Structure determination.** Solve the structure of the KT2440 XdhAB complex (cryo-EM or X-ray) to confirm quaternary arrangement, the [2Fe-2S]→FAD wiring in XdhA, and to enable direct comparison with the *R. capsulatus* XDH and *P. putida* 86 quinoline 2-oxidoreductase structures.
5. **XdhC dependence test.** Compare holoenzyme activity produced with and without co-expressed XdhC to directly demonstrate that KT2440 XdhC is required for maturation of active XDH, as shown for the *R. capsulatus* ortholog.
6. **Substrate-specificity profiling.** Assay the reconstituted enzyme against a panel of purines and aromatic aldehydes (and methylxanthine intermediates such as 1- and 3-methylxanthine) to define the substrate range and any competitive inhibition by uric acid, benchmarking against the strain-86 and strain-40 data.
7. **Transcriptional/inducibility analysis.** Quantify xdhA operon expression (RT-qPCR / RNA-seq) under purine vs. non-purine nitrogen sources to confirm substrate-inducibility (analogous to the ~65-fold hypoxanthine induction reported for strain 86), map operon boundaries with xdhB, xdhC, and guaD, and test regulation by the adjacent GntR-family regulator (PP_4277).

Conclusion

The gene symbol "xdhA" is **not** ambiguous for this target: all lines of evidence — the UniProt description, the InterPro/Pfam domain architecture (two 2Fe-2S ferredoxin domains + PCMH FAD-binding domain), eggNOG COG4630, and KEGG K13481 — agree that **Q88F21 / PP_4278 encodes the FAD- and 2×[2Fe-2S]-containing small (electron-transfer) subunit of a two-component bacterial xanthine dehydrogenase (EC 1.17.1.4) in *P. putida* KT2440**. Paired with the molybdenum-cofactor catalytic subunit XdhB (PP_4279), the enzyme oxidizes hypoxanthine → xanthine → uric acid with NAD⁺ as the preferred electron acceptor; XdhA's specific job is to relay electrons from the molybdenum center through its [2Fe-2S] clusters to FAD and thence to NAD⁺. The enzyme is a soluble cytoplasmic complex that functions in the purine-degradation pathway enabling purines to serve as nitrogen sources, requires the operonic accessory factor XdhC for molybdenum-cofactor maturation, and is fed by the adjacent guanine deaminase (GuaD).